

A Lava Well of Reflexivity: Exploring Speculative Ambient Media

Ting Han Daniel Chen
Play Design Lab
Taiwan
daniel@playdesignlab.com



Figure 1: Snapshots of the installation A Lava Well of Reflexivity.

Abstract

This work explores the idea of speculative ambient media, which integrates reflexive and speculative elements into environmental objects. Unlike traditional ambient media that provides background information, speculative ambient media subtly promotes reflexivity on societal and environmental issues. Through the case, we illustrate how such media might inspire contemplation without being intrusive. The installation features slow-changing projections from plastic waste to implicate and raise environmental awareness and challenge perceptions of human impact on nature. Speculative ambient media enhances environmental consciousness and fosters dialogue. Future research will investigate its application in various contexts to further assess its potential for stimulating discussion and reflection.

CCS Concepts

• **Applied computing** → Arts and humanities; Media arts.

Keywords

Ambient media, Speculative design, Intervention, Light installation, Critical design

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1 Introduction

Traditional ambient media (Gellersen et al., 1999) focuses on delivering background information, whereas our concept emphasizes embedding reflexive and speculative elements within objects (Fig 1). This inherent meaning aligns with the goals of speculative design, which explores the potential impacts of today's decisions on tomorrow's world, encouraging a deeper consideration of the long-term consequences of our actions (Wong and Khovanskaya, 2018). Through ambient forms, these implications encourage critical reflection and awareness without being overt, similar to how everyday products made from recycled materials serve as gentle reminders of sustainability.

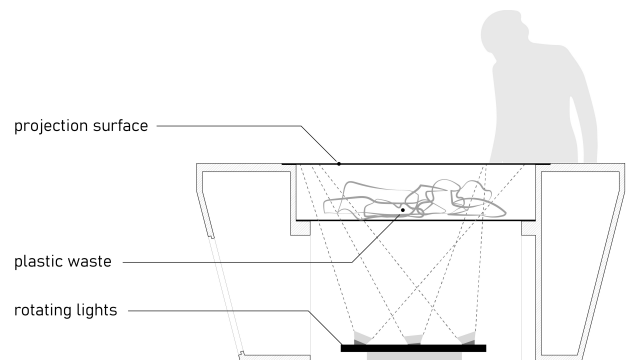


Figure 2: Interior deployment within the installation.

2 Speculative Ambient Media

Speculative ambient media differs from traditional forms by embedding reflexive and speculative characteristics in environmental objects. Key features include:

Subtle Presence. It quietly exists in the environment, allowing for optional engagement. Its speculative nature becomes apparent upon closer inspection, creating an ambient atmosphere.

Gentle Engagement. Unlike conventional speculative design, which often presents issues directly, speculative ambient media invites discovery and exploration. It remains unobtrusive until the observer chooses to engage, offering a more approachable interaction.

Speculative Insight. Rather than providing personalized or individualized “information,” it embeds “insight” that transcends the immediate personal level and encourages reflections on broader societal, political, or environmental scenarios.

Enhanced Impact. When the observer comprehends the implications, its speculative qualities become impactful. The contrast between its usual ambient state and the moment of focused attention creates a strong psychological effect, emphasizing its intrinsic critical message.

3 Case Overview

This case describes a speculative ambient media art installation, designed following the principles of slow technology (Hallnäs and Redström, 2001), for the central hall of the Good Earth Pavilion at the New Taipei City Gold Museum. As shown in Figure 2, the top projection surface of the installation is a frosted glass panel adorned with lava-like mineral patterns. The patterns are created by a set of slow-moving projection lights rotating at approximately 1.5 degrees per second. The visual aesthetic of the lava-like patterns varies subtly when observed at different times (Fig 3). These patterns symbolize the changes in the earth’s environment over time, reflecting how environmental parameters dynamically and continuously interact, leading to corresponding transformations in the surroundings. The slow rotation makes these continuous changes difficult to notice without careful observation, subtly highlighting the gradual nature of environmental shifts.

A deeper interpretation emerges when viewers recognize that the lava-like patterns are projections formed from plastic waste, where the installation includes overlapping and carelessly positioned plastic bags and packaging sourced from local mountains (Fig 4). This artistic choice provides a critical perspective on the relationship between humans, artificial waste, and natural resources. It invites viewers to contemplate the implications of future natural formations derived from accumulated waste and the creation of new aesthetics from synthetic materials. By challenging conventional perceptions of the human-environment relationship, the installation subtly prompts viewers to reconsider the critical issues we aim to address.

4 Discussion and Future Work

Speculative ambient media gently encourages reflection on societal and environmental issues, setting it apart from traditional ambient media, which often focuses on personal or practical information. By introducing unexpected moments that invite contemplation, it has the potential to spark conversations and exchanges of ideas. However, its subtle nature may reduce its ability to prompt immediate reflection, as some viewers may overlook or not engage with the prompts. To enhance its critical impact, future projects could

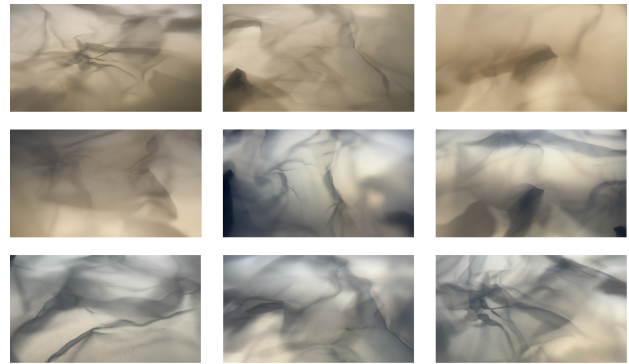


Figure 3: The lava-like patterns shaped by different lighting parameters—such as angles, positions, and brightness—at various times represent the Earth’s transformation under constantly changing environmental factors.



Figure 4: Plastic waste, a primary material component, within the installation.

integrate more interactive features or establish stronger links to tangible environmental consequences.

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